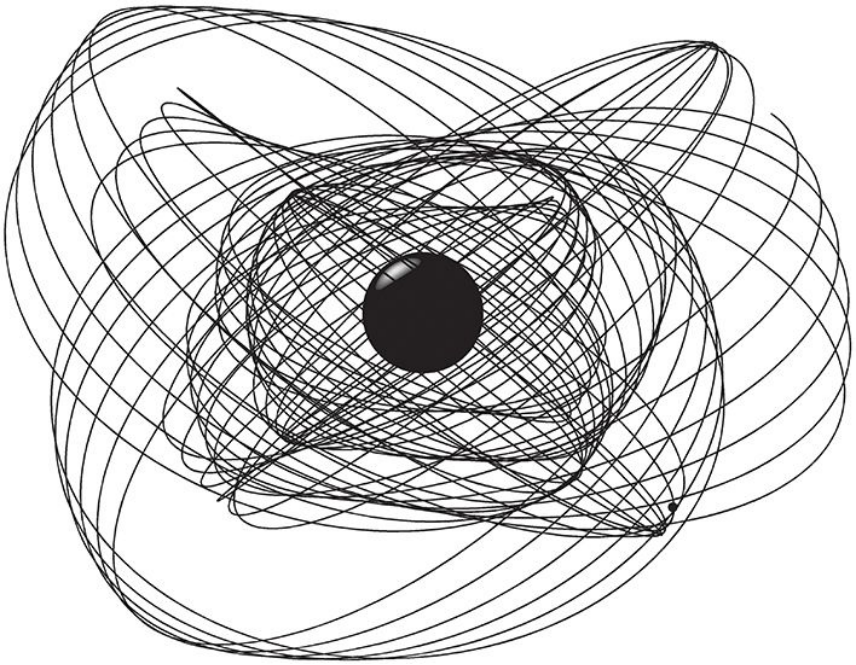


Fig. 7.6. A single orbit of a spacecraft or planet or star around a gigantic, fast-spinning black hole such as Gargantua. *[From a simulation by Steve Drasco.]*

I'll let *you* imagine how an ultra-advanced civilization might use such complex orbits. In my science interpretations of the movie, for simplicity I mostly eschew them and focus primarily on circular, equatorial orbits (those of the parked *Endurance*, Miller's planet, and the critical orbit), and on simple trajectories for the *Endurance* as it travels from one circular equatorial orbit to another. An exception is the orbit of Mann's planet, discussed in Chapter 19.

NASA's Gravitational Slingshots in the Solar System

Let's return from the world of the possible (what the laws of physics allow) to hard-nosed, real-life gravitational slingshots in the comfy confines of our solar system (what humans have actually achieved as of 2014).